

Dylan Pierre

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PROFESSIONAL SUMMARY

Computer Science graduate with experience across backend development, systems programming, machine learning, and software security. Seeking a Software Engineering role focused on building software that is reliable, scalable, performant, and secure.

EXPERIENCE

Surge AI - DataAnnotation

May 2024 – Present

Code Quality Analyst

Contract

- Contributed to training and evaluation datasets, improving **frontier AI models** developed by Anthropic and OpenAI.
- Worked directly with **Claude Code** and **GitHub Copilot** to drive codebase improvements and evaluate AI-generated code for **correctness**, **security**, and **performance**.
- Documented evaluation results and technical findings to support **model improvements** and alignment with technical requirements.

TECHNICAL PROJECTS

Checkpoint | Go, PostgreSQL, Redis, Nginx, Docker, Git/GitHub, RESTful API | [\[GitHub\]](#)

- Built a concurrent authentication and authorization backend in **Go**, using **PostgreSQL** and **Redis** to manage persistence and state.
- Containerized the service with **Docker** and used **Nginx** to load balance separate app instances.
- Load tested the service under concurrent traffic, achieving **12,100+ requests/second** with **100% success** and **23.3ms tail latency** over **121,000+ requests**.
- Hardened the service against credential theft, token forgery, and brute-force attacks using OWASP standard **Argon2id** hashing parameters and **PASETO** tokens.

Backbone | Python, PyTorch, Machine Learning, Computer Vision, CT | [\[Link\]](#)

- Designed a machine learning pipeline for cervical spine fracture detection in CT imaging, using **PyTorch** for detection and **pydicom/OpenCV** for DICOM preprocessing.
- Built a data pipeline containing **14,434 labeled CT slices** from the RSNA 2022 dataset to train and fine-tune a **YOLO** model.
- Collaborated on an Agile team under a fast-paced timeline, running weekly stand-ups to deliver a working system.

C-Like Language Interpreter | C++, Interpreters, CMake | [\[GitHub\]](#)

- Built a language interpreter in **C++**, implementing the tokenizer, recursive-descent parser, abstract syntax tree, symbol table, and evaluator with **zero external dependencies**.
- Interprets typed variables, arrays, functions, and logical control flow (**if/else**, **for**, **while**) with scope-aware declarations and line-numbered error diagnostics.
- Deployed the project to a remote **Linux** server over **SSH** and configured the C++ build with **CMake**.
- Validated the implementation with **25 test programs**, individually testing the tokenizer, parser, symbol table, and evaluator.

Portfolio Website | React, TypeScript, HTML, CSS, Vite, npm | [\[Site\]](#)

- Built a personal portfolio deployed with Vercel, using **React**, **TypeScript**, **Vite**, and **HTML/CSS**.

EDUCATION

Sonoma State University

3.5 GPA

Bachelor of Science in Computer Science

2025

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Algorithm Analysis, Operating Systems, Database Systems, Computer Networks, Programming Languages, Software Engineering, Machine Learning, Artificial Intelligence

CERTIFICATIONS

CompTIA Security+ (April 2026), **Google Cybersecurity Professional**